

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2019 - 2020)

NEET(UG)
PART TEST-3
16-02-2020

PRE-MEDICAL : SPARK COURSE (PHASE - I)*This Booklet contains 24 pages.***Do not open this Test Booklet until you are asked to do so.****Read carefully the Instructions on the Back Cover of this Test Booklet.****Important Instructions :**

1. On the Answer Sheet, fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are 720.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
4. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
5. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

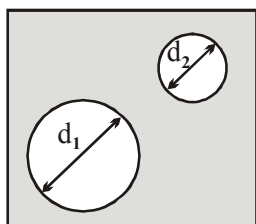
: in words _____

Centre of Examination (in Capitals) : _____

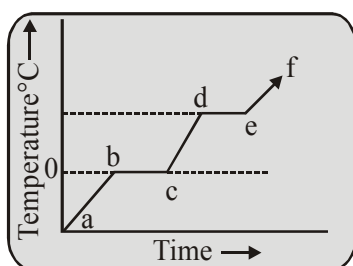
Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in Pre-Medical 2020

1. Two holes of unequal diameters d_1 and d_2 ($d_1 > d_2$) are cut in a metal sheet. If the sheet is heated :-



- (1) Both d_1 and d_2 will decrease
 (2) Both d_1 and d_2 will increase
 (3) d_1 will increase, d_2 will decrease
 (4) d_1 will decrease, d_2 will increase
2. Coefficient of linear expansion of brass and steel rods are α_1 and α_2 . Lengths of brass and steel rods are l_1 and l_2 respectively. If $(l_2 - l_1)$ is maintained same at all temperatures, which one of the following relations holds good?
- (1) $\alpha_1 l_2 = \alpha_2 l_1$ (2) $\alpha_1 l_2^2 = \alpha_2 l_1^2$
 (3) $\alpha_1^2 l_2 = \alpha_2^2 l_1$ (4) $\alpha_1 l_1 = \alpha_2 l_2$
3. The following figure represents the temperature *versus* time plot for a given amount of a substance when heat energy is supplied to it at a fixed rate and at a constant pressure.



Which part of the above plot represent a state change?

- (1) a to b and e to f (2) b to c and c to d
 (3) d to e and e to f (4) b to c and d to e

4. Two temperature scales A and B are related by $\frac{A - 42}{110} = \frac{B - 72}{220}$. At which temperature two scales have the same readings?

(1) -42° (2) -72° (3) $+12^\circ$ (4) -40°

5. Three liquids with masses m_1 , m_2 , m_3 are thoroughly mixed. If their specific heats are c_1 , c_2 , c_3 and their temperature T_1 , T_2 , T_3 respectively, then the temperature of the mixture is :-

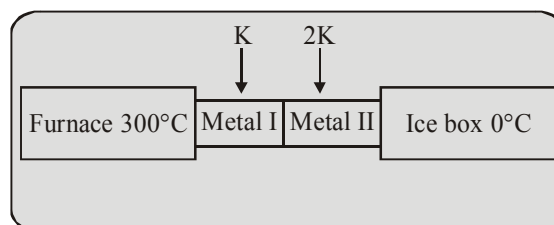
(1) $\frac{c_1 T_1 + c_2 T_2 + c_3 T_3}{m_1 c_1 + m_2 c_2 + m_3 c_3}$

(2) $\frac{m_1 c_1 T_1 + m_2 c_2 T_2 + m_3 c_3 T_3}{m_1 c_1 + m_2 c_2 + m_3 c_3}$

(3) $\frac{m_1 c_1 T_1 + m_2 c_2 T_2 + m_3 c_3 T_3}{m_1 T_1 + m_2 T_2 + m_3 T_3}$

(4) $\frac{m_1 T_1 + m_2 T_2 + m_3 T_3}{c_1 T_1 + c_2 T_2 + c_3 T_3}$

6. In the diagram, a system of two metals of equal lengths and of same cross-sectional area are joined together.

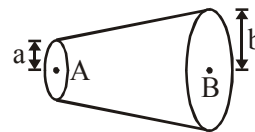


The coefficient of thermal conductivities of the metals are K and 2K, respectively. If the furnace temperature at one end is 300°C and ice box temperature at the other end is 0°C , then the junction temperature is :

- (1) 100°C (2) 125°C
 (3) 150°C (4) 200°C

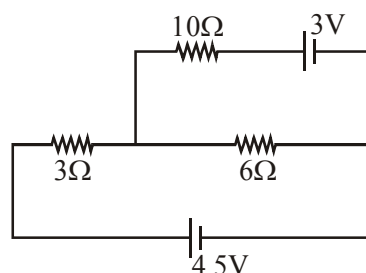
7. A Carnot engine operates with source at 127°C and sink at 27°C . If the source supplies 40 kJ heat energy, the work done by the engine is :-
 (1) 30 kJ (2) 10 kJ (3) 4 kJ (4) 1 kJ
8. Two mole of oxygen is mixed with eight mole of helium. The effective molar specific heat of the mixture at constant volume is :-
 (1) 1.3 R (2) 1.4 R (3) 1.7 R (4) 1.9 R
9. The velocities of three molecules are $3v$, $4v$ and $5v$ respectively. Their rms speed will be :-
 (1) $\sqrt{\frac{50}{3}}v$ (2) $\sqrt{\frac{5}{2}}v$ (3) $\frac{7}{2}v$ (4) $\frac{5}{2}v$
10. A body cools in 7 min from 60°C to 40°C . What time (in min) does it take to cool from 40°C to 28°C , if surrounding temperature is 10°C ? (Assume, Newton's law of cooling is valid.)
 (1) 3.5 (2) 12 (3) 7 (4) 14
11. If temperature of a black body increases from -73°C to 327°C , then ratio of emissive power at these two temperature is :-
 (1) 27 : 1 (2) 81 : 1 (3) 1 : 27 (4) 1 : 81
12. Two containers of equal volume contain the same gas at pressure P_1 and P_2 and absolute temperature T_1 and T_2 , respectively. On joining the vessels, the gas reaches a common pressure P and common temperature T . The ratio P / T is equal to :-
 (1) $\frac{P_1}{T_1} + \frac{P_2}{T_2}$ (2) $\frac{P_1 T_1 + P_2 T_2}{(T_1 + T_2)^2}$
 (3) $\frac{P_1 T_2 + P_2 T_1}{(T_1 + T_2)^2}$ (4) $\frac{P_1}{2T_1} + \frac{P_2}{2T_2}$
13. A monoatomic gas is suddenly compressed to $(1/8)^{\text{th}}$ of its initial volume adiabatically. The ratio of its final pressure to the initial pressure is (given the ratio of the specific heats of the given gas to be $5/3$) :-
 (1) 32 (2) $40/3$ (3) $24/5$ (4) 8
14. Current flowing through a conducting wire is given by $I = (1 + 2t)$, where 't' is in seconds and current I is in amp. The charge (in coulombs) flow through the resistor in the interval from $t = 0$ to $t = 1$ sec is
 (1) 3 (2) 2 (3) 1 (4) 0

15. An electric current passes through Non-uniform cross-section wire made of homogeneous and isotropic material. If the J_A & J_B be the current densities and E_A and E_B be the electric field intensities at A and B respectively, then



- (1) $J_A > J_B$; $E_A > E_B$
 (2) $J_A > J_B$; $E_A < E_B$
 (3) $J_A < J_B$; $E_A > E_B$
 (4) $J_A < J_B$; $E_A < E_B$

16. The current through the $10\ \Omega$ resistor is



- (1) zero (2) 1 A (3) 2 A (4) 3 A

17. A rectangular carbon block has dimensions $1.0\text{ cm} \times 1.0\text{ cm} \times 50\text{ cm}$. The resistance measured between the two square ends is. (Resistivity of carbon is $3.5 \times 10^{-5}\ \Omega\text{-m}$)

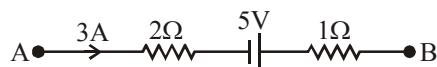
- (1) $17.5\ \Omega$ (2) $0.175\ \Omega$
 (3) $1.75\ \Omega$ (4) $\frac{1000}{175}\ \Omega$

18.
 (a) (b)
 (c) (d)

In which of the above cells, the potential difference between the terminals of a cell less than its emf.

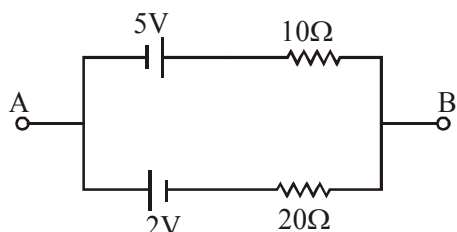
- (1) a (2) b (3) c (4) d

19. The length of a given cylindrical wire is increased by 50% of its initial length, the percentage increase in the resistance of the wire is :-
 (1) 300% (2) 150% (3) 125% (4) 100%
20. The potential difference ($V_B - V_A$) between the points A & B in the given figure is :-



- (1) +6V (2) +9V
 (3) +4 Volt (4) -4 Volt

21.



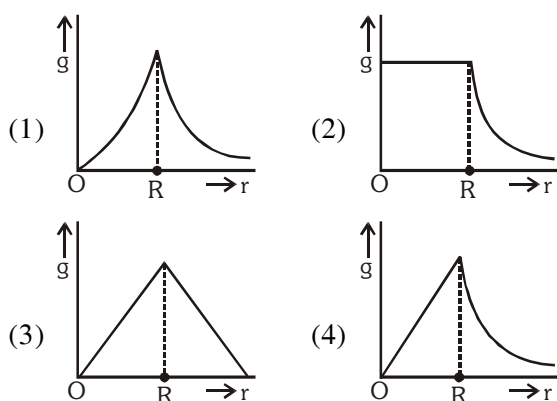
In the following circuit the value of ($V_A - V_B$) is :

- (1) $-\frac{2}{3}$ volt (2) 4 volt
 (3) $-\frac{8}{3}$ volt (4) 7 volt

22. A body weighs W newton at the surface of the earth. Its weight at a height equal to half the radius of the earth will be :

- (1) $\frac{W}{2}$ (2) $\frac{2W}{3}$ (3) $\frac{4W}{9}$ (4) $\frac{W}{4}$

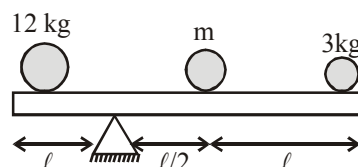
23. Starting from the centre of the earth having radius R , the variation of g (acceleration due to gravity) is shown by :-



24. The acceleration due to gravity at a height 1 km above the earth is the same as at a depth d below the surface of earth. Then :-

- (1) $d = 1$ km (2) $d = \frac{3}{2}$ km
 (3) $d = 2$ km (4) $d = \frac{1}{2}$ km

25. For equilibrium of the system, value of mass m should be :



- (1) 9 kg (2) 15 kg (3) 21 kg (4) 1 kg

26. A wheel having moment of inertia 4 kg-m^2 about its axis rotates at rate of 240 rotation per minute about it. The torque which can stop the rotation of the wheel in one minute is :

- (1) $\frac{5\pi}{7} \text{ Nm}$ (2) $\frac{8\pi}{15} \text{ Nm}$
 (3) $\frac{2\pi}{9} \text{ Nm}$ (4) $\frac{3\pi}{7} \text{ Nm}$

27. If radius of earth suddenly contracts by half of its initial value keeping mass constant, then what will be the time period of rotation of earth about its axis:
 (1) 6 hr (2) 12 hr (3) 3 hr (4) 4 hr

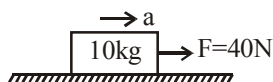
28. A solid cylinder rolls down without slipping on an inclined plane of angle of inclination θ . What should be the minimum value of coefficient of friction between cylinder and inclined surface for pure rolling motion :

- (1) $\frac{1}{3} \tan \theta$ (2) $\tan \theta$ (3) $\frac{2}{3} \tan \theta$ (4) $3 \tan \theta$

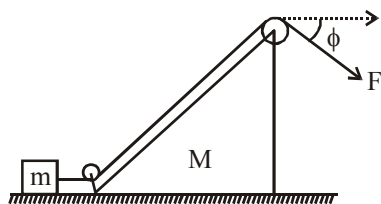
29. A thin circular ring first slips down a smooth incline then rolls down a rough incline of identical geometry from same height. Ratio of time taken in the two motion is:

- (1) $\frac{1}{2}$ (2) 1 (3) $\frac{1}{\sqrt{2}}$ (4) $\frac{1}{4}$

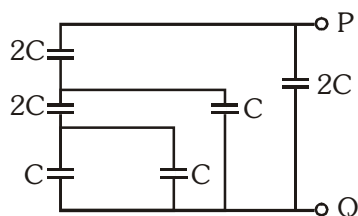
30. A block of mass 10kg moving with acceleration 2 m/s^2 on horizontal rough surface is shown in figure. The value of coefficient of kinetic friction is:



- (1) 0.2 (2) 0.4
(3) 0.5 (4) 0.1
31. In the given arrangement pulleys and string are massless and frictionless and all surfaces are smooth. Find the magnitude of acceleration of wedge of mass M.



- (1) $\frac{F \cos \phi}{M}$ (2) $\frac{F(1 - \cos \phi)}{M}$
(3) $\frac{F}{M + m}$ (4) $\frac{F \cos \phi}{M + m}$
32. The value of equivalent capacitance of the combination shown in figure, between the points P and Q is :

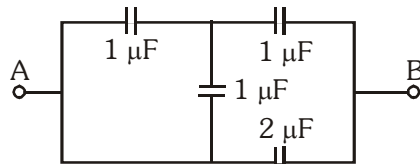


- (1) 3C (2) 2C
(3) C (4) C/3
33. Half of the space between a parallel plate capacitor is filled with a medium of dielectric constant K parallel to the plates. If initially the capacity was C, then the new capacity will be :
- (1) $2KC/(1+K)$ (2) $C(K+1)/2$
(3) $CK/(1+K)$ (4) KC

34. Two identical parallel plate capacitors are connected in series and then joined with a battery of 100 V. A sheet of dielectric constant 4.0 is inserted between the plates of second capacitor. The potential difference across the capacitors will be respectively :

- (1) 50 V, 50 V (2) 80 V, 20 V
(3) 20 V, 80 V (4) 75 V, 25 V

35. The equivalent capacitance between points A and B of the circuit shown will be :

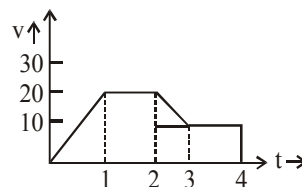


- (1) $\frac{2}{3} \mu\text{F}$ (2) $\frac{5}{3} \mu\text{F}$
(3) $\frac{8}{3} \mu\text{F}$ (4) $\frac{7}{3} \mu\text{F}$

36. A lift performs the first part of ascent with uniform acceleration a and the remainder with uniform retardation 2a. The lift starts from rest and finally comes to rest. If t is the time of ascent. Find the height ascended by lift.

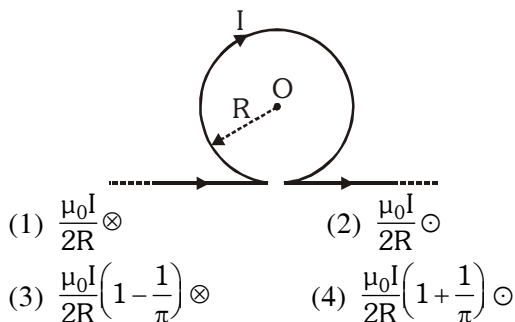
- (1) $\frac{at^2}{4}$ (2) $\frac{at^2}{3}$
(3) $\frac{at^2}{2}$ (4) $\frac{at^2}{8}$

37. The variation of velocity of a particle with time moving along a straight line is illustrated in the following figure. The distance travelled by the particle is four seconds is :-



- (1) 60 m (2) 55 m (3) 25 m (4) 30 m

38. Magnetic field at point O will be :-



39. If radius of coil becomes two times and current becomes half then magnetic field at centre of the coil will be :-

- (1) Two times (2) Four times
 (3) Half (4) One fourth

40. Mean radius of a toroid is 10 cm and number of turns are 500. If current flowing through it is 0.1 ampere then value of magnetic field (in tesla) for toroid :-

- (1) 10^{-2} (2) 10^{-5}
 (3) 10^{-3} (4) 10^{-4}

41. A charge having q/m equal to 10^8 C/kg and with velocity 3×10^5 m/s enters into a uniform magnetic field $B = 0.3$ tesla at an angle 30° with direction of field. Then radius will be :-

- (1) 0.01 cm (2) 0.5 cm (3) 1 cm (4) 2 cm

42. An electric field E and a magnetic field B applied on a proton which moves with velocity v , it goes undeflected through the region if :-

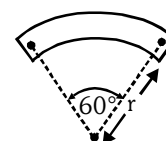
- (1) $E \perp B$
 (2) E is parallel to v and perpendicular to B
 (3) E , B and v all three mutually perpendicular to each other and $v = E/B$
 (4) E and B both are parallel but perpendicular to v

43. P, Q and R long parallel straight wires in air, carrying currents as shown. The direction of resultant force on R is :-

- (1) towards left
 (2) towards right
 (3) the same as that of current in Q
 (4) perpendicular to plane of paper

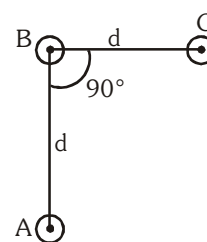


44. A bar magnet of length ' l ' and magnetic dipole moment ' M ' is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be :-



- (1) $\frac{M}{2}$ (2) M
 (3) $\frac{3}{\pi} M$ (4) $\frac{2}{\pi} M$

45. An arrangement of three parallel straight wires placed perpendicular to plane of paper carrying same current ' I ' along the same direction is shown in fig. Magnitude of force per unit length on the middle wire 'B' is given by :-

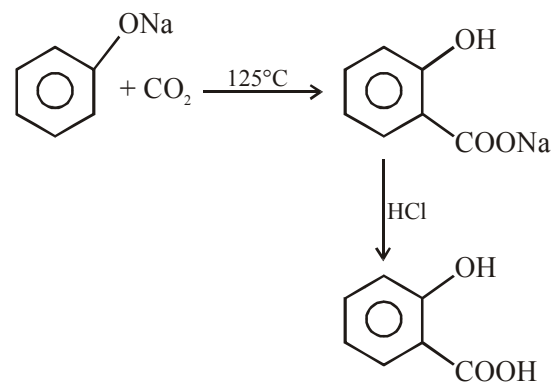
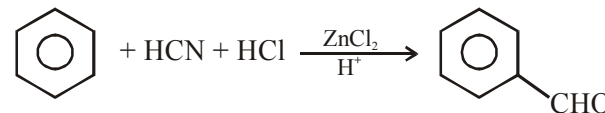
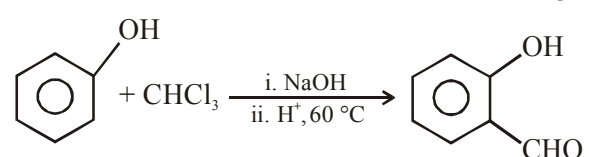
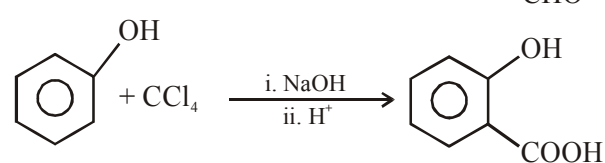


- (1) $\frac{2\mu_0 I^2}{\pi d}$ (2) $\frac{\sqrt{2}\mu_0 I^2}{\pi d}$
 (3) $\frac{\mu_0 I^2}{\sqrt{2}\pi d}$ (4) $\frac{\mu_0 I^2}{2\pi d}$

46. Formation of but-2-ene as major product by dehydration of butan-2-ol is according to _____.

- (1) Markownikoff's rule
- (2) Saytzeff rule
- (3) Peroxide effect
- (4) Anti-Markownikoff's rule

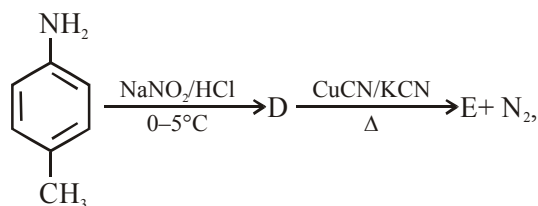
47. Which of the following reaction is Kolbe's reaction ?

- (1) 
- (2) 
- (3) 
- (4) 

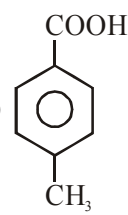
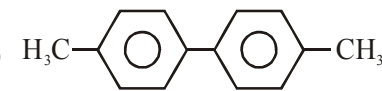
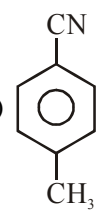
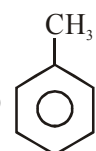
48. Pentan-3-one can be obtained by dry distillation of _____.

- (1) calcium acetate and calcium formate
- (2) calcium propionate
- (3) calcium acetate
- (4) calcium propionate and calcium formate

49. In the reaction



the product E is _____.

- (1) 
- (2) 
- (3) 
- (4) 

50. The reaction of aniline with chloroform under alkaline conditions leads to the formation of _____.

- (1) phenyl cyanide
- (2) phenyl isonitrile
- (3) phenyl cyanate
- (4) phenyl isocyanate

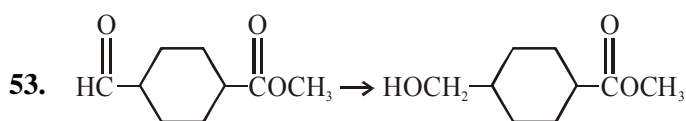
51. In the formation of Ph – CHO, following reactions are used. Match them CORRECTLY.

Reaction	Reagent
[A] Rosenmund	(i) CO+HCl
[B] Gattermann-Koch	(ii) CrO ₂ Cl ₂
[C] Etard	(iii) SnCl ₂ /HCl
[D] Stephen	(iv) Pd/BaSO ₄

- (1) [A]-(iv), [B]-(iii), [C]-(iii), [D]-(i)
 (2) [A]-(iv), [B]-(i), [C]-(ii), [D]-(iii)
 (3) [A]-(iv), [B]-(ii), [C]-(iii), [D]-(i)
 (4) [A]-(iv), [B]-(iii), [C]-(i), [D]-(ii)

52. Ethyl methyl ether on treating with cold concentrated HI gives_____.

- (1) C₂H₅OH (2) CH₃I
 (3) C₂H₅I + CH₃OH (4) CH₃I + C₂H₅OH



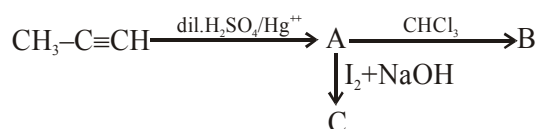
The reduction can be achieved by using_____.

- (1) NaBH₄ (2) LiAlH₄
 (3) NaOH (4) all of these

54. The most suitable reagent for the conversion of R-CH₂-OH → R – CHO is_____.

- (1) KMnO₄
 (2) K₂Cr₂O₇
 (3) CrO₃
 (4) PCC (Pyridinium chlorochromate)

55. In the reaction sequence :-

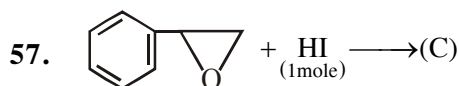


A, B and C are given by the set :-

- (1) CH₃CH₂CHO, CH₃CH₂CH₂Cl, CHI₃
 (2) CH₃COCH₃, CCl₃-C(=O)-CH₃, CHI₃
 (3) CH₃COCH₃, CCl₃-C(OH)(CH₃)₂, CHI₃
 (4) CH₃CH₂CHO, CCl₃-CH(OH)-CH₂-CH₃, CHI₃

56. For the electrophilic substitution reaction involving sulphonation, which of the following sequence regarding the rate of reaction is true?

- (1) k_{C₆H₆} > k_{C₆D₆} > k_{C₆T₆}
 (2) k_{C₆H₆} < k_{C₆D₆} < k_{C₆T₆}
 (3) k_{C₆H₆} = k_{C₆D₆} = k_{C₆T₆}
 (4) k_{C₆H₆} > k_{C₆D₆} < k_{C₆T₆}

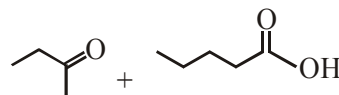
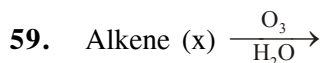


(C) is :-

- (1)
- (2)
- (3)
- (4) None of these

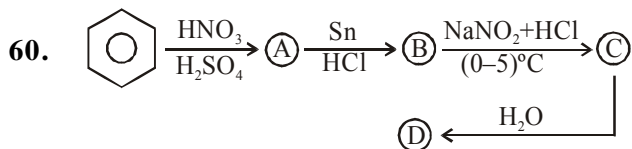
58. To prepare propan-2-ol from methyl magnesium bromide, the other reagent required is_____.

- (1) HCHO (2) CH₃CHO
 (3) C₂H₅OH (4) CH₃COCH₃

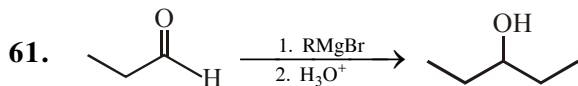
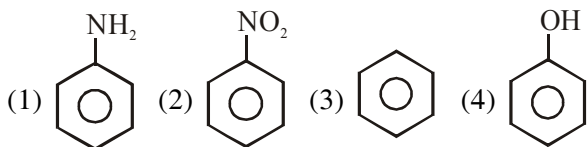


x is :-

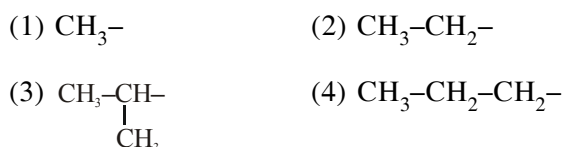
- (1)
- (2)
- (3)
- (4)



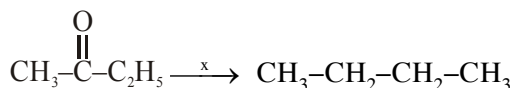
'D' is



R is :-



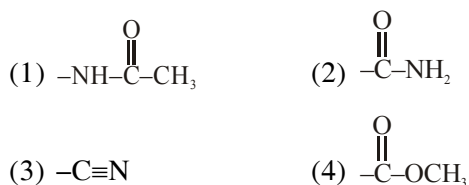
62. In given reaction :



'X' is :-

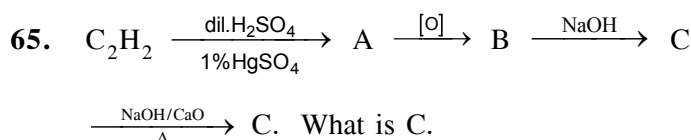


63. Which of the following group is ortho para director :-



64. Which of the following is not correctly matched:-

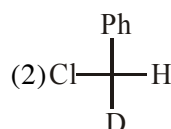
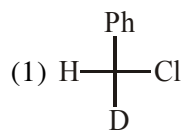
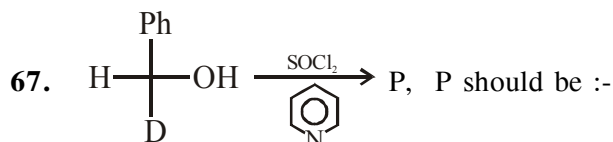
- (1) Sorbic acid $\rightarrow$ Food preservative
(2) Sodium stearate $\rightarrow$ Soap
(3) Alitame $\rightarrow$ Sweetner
(4) Sodium benzoate $\rightarrow$ Detergents



- (1) CH_4 (2) CH_3-CH_3
(3) C_3H_8 (4) C_4H_{10}

66. Which is best leaving group ?

- (1) $\text{H}_3\text{C}-\text{O}^\ominus$ (2) $\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{S}}-\text{O}^\ominus$
(3) $\text{H}_3\text{C}-\text{CH}_2^\ominus$ (4) $\text{H}_3\text{C}-\text{NH}^\ominus$



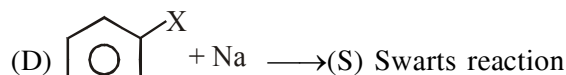
- (3) Both 1 & 2 in equal proportions
(4) None of these

68. Which one of the following is least reactive with water



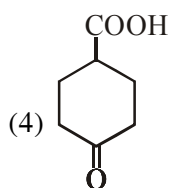
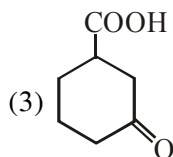
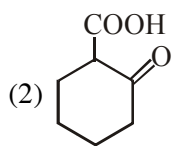
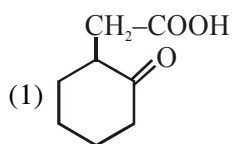
69. Choose the correct option ?

- (A) $\text{R}-\text{OH} + \text{SOCl}_2 \longrightarrow (\text{P})$ Finkelstein reaction
(B) $\text{R}-\text{X}(\text{Cl}, \text{Br}) + \text{NaI} \longrightarrow (\text{Q})$ Wurtz fittig reaction
(C) $\text{R}-\text{X}(\text{Cl}, \text{Br}) + \text{AgF} \longrightarrow (\text{R})$ Darzen process

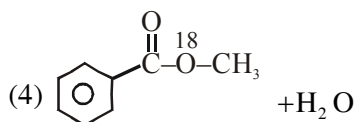
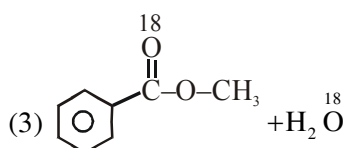
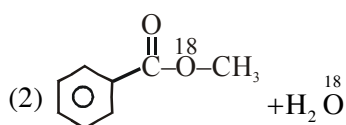
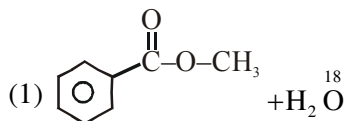
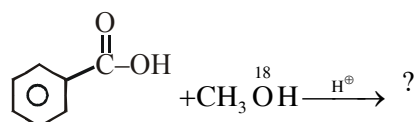


- (1) $\text{A} \rightarrow \text{P}$ (2) $\text{B} \rightarrow \text{R}$
(3) $\text{C} \rightarrow \text{S}$ (4) $\text{D} \rightarrow \text{Q}$

70. Which of following is maximum reactive for decarboxylation :-



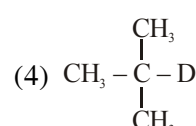
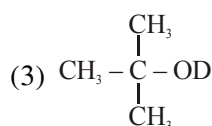
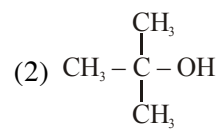
71. What is the product obtained from the following reaction ?



72. 1° amine and 2° Amine can be differentiated by :-

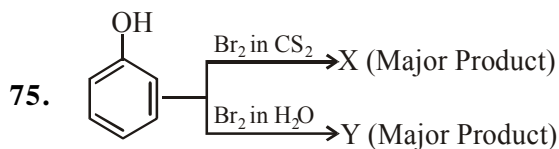
- (1) $CHCl_3/Alc.KOH$
- (2) $NaNO_2/HCl$
- (3) $PhSO_2Cl$ then $NaOH$
- (4) All of these

73. $(Me)_3CMgCl$ on reaction with D_2O produces ?

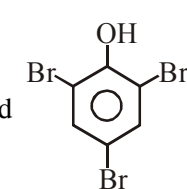
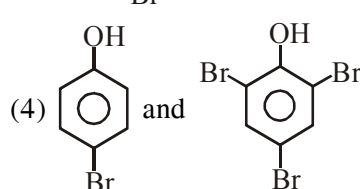
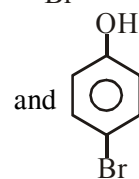
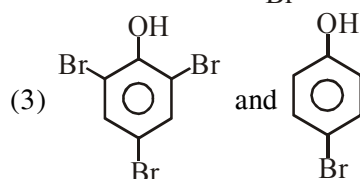
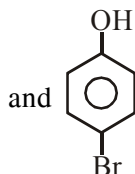
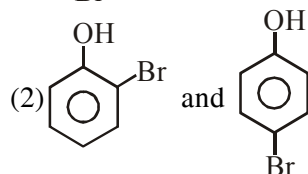
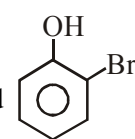
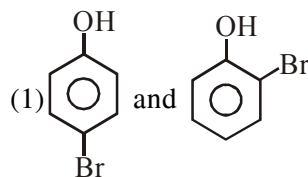


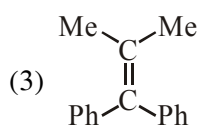
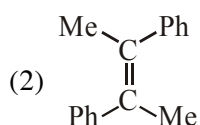
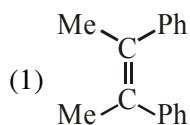
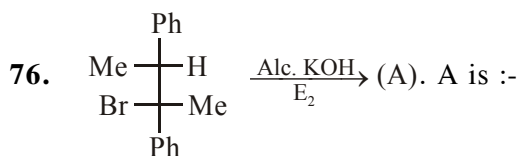
74. Which of the following compound can be best prepared by wurtz-reaction ?

- (1) Iso butane
- (2) Neo pentane
- (3) Butane
- (4) Methane



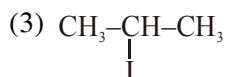
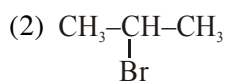
What is X and Y in above reaction respectively :-



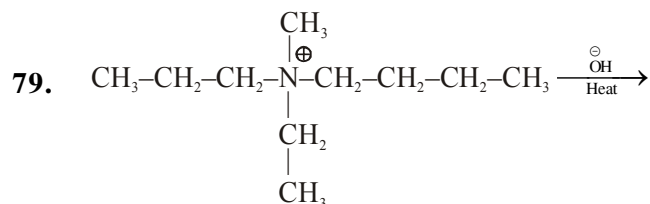
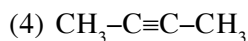
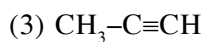
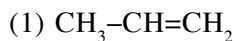


(4) None is correct

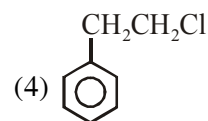
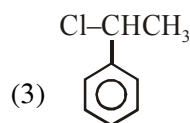
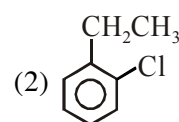
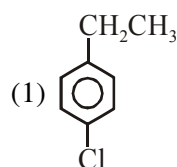
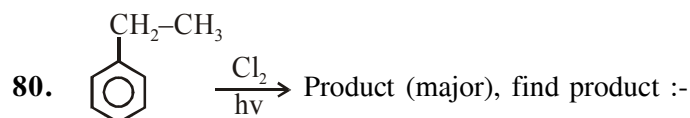
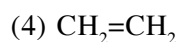
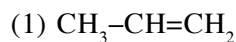
77. Which of the following alkyl halides gives the fastest $\text{S}_{\text{N}}1$ reaction :-



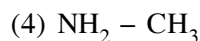
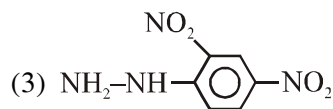
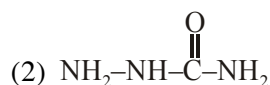
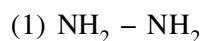
78. Which of the following give white ppt with Tollen's reagent :-



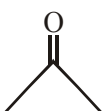
Major product is :-



81. $\text{CH}_3\text{CHO} \xrightarrow{\text{Reagent}}$ Product (yellow or orange ppt) (Reagent is also used to identify carbonyl compound). Find reagent :-



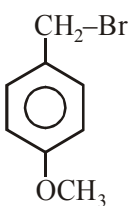
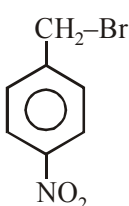
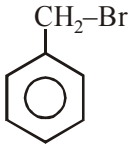
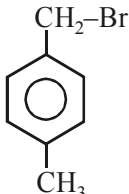
82. Which compound can form stable hydrate :-

- (1) HCHO (2) 
- (3) Cl₃CCHO (4) All

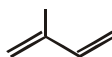
83. Gabriel phthalimide reaction is used for the preparation of :-

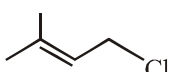
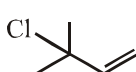
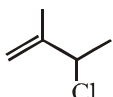
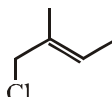
- (1) Primary aromatic amines
(2) Secondary amines
(3) Aliphatic primary amine
(4) Tertiary amine

84. Arrange the following in reactivity order towards S_N¹ :-

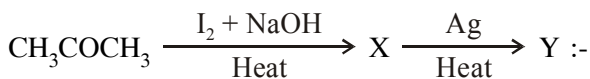
- (1)  (2) 
- (3)  (4) 

- (1) 1 > 4 > 3 > 2 (2) 1 > 3 > 4 > 2
(3) 3 > 1 > 2 > 4 (4) 2 > 1 > 3 > 4

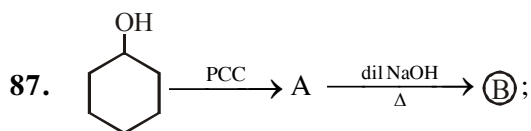
85.  $\xrightarrow{\text{HCl (1 mole)}}$ Major product :-

- (1)  (2) 
- (3)  (4) 

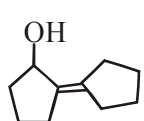
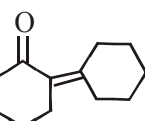
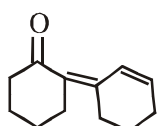
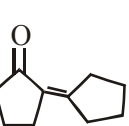
86. Identify (X) and (Y) in the given reaction.



- (1) X = CHI₃, Y = CH₃CHO
(2) X = CHI₃, Y = CH₃OH
(3) X = CHI₃, Y = CH ≡ CH
(4) X = CH₃COCl, Y = CH₂ = CH₂



final product 'B' will be :

- (1)  (2) 
- (3)  (4) 

88. Consider the following reaction



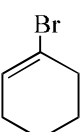
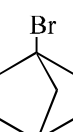
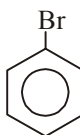
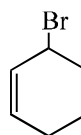
The product A is :-

- (1) PhCOCH₃ (2) Ph-Cl
(3) PhCHO (4) Ph-OH

89. Aniline on treatment with bromine water yields white precipitate of :-

- (1) o-Bromoaniline
(2) 2,4,6-Tribromo aniline
(3) p-Bromoaniline
(4) m-bromoaniline

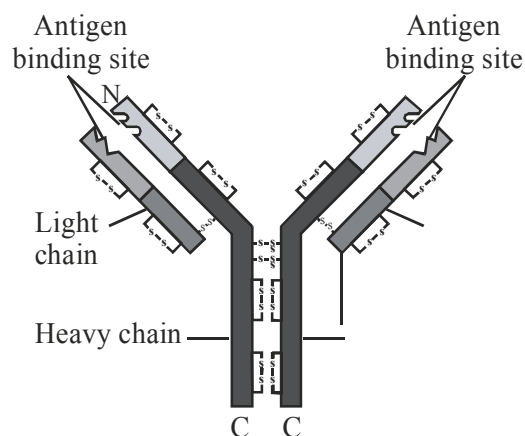
90. S_N2 reaction will be negligible in

- (I)  (II) 
- (III)  (IV) 
- (1) Only IV (2) I, II, III
(3) I, III (4) Only II

91. Where immature lymphocytes differentiate into antigen sensitive lymphocyte?

- (1) MALT
- (2) Primary lymphoid organs
- (3) Secondary lymphoid organs
- (4) Lymph nodes

92. Which of the following is/are correct for given diagram :-



- (a) Produced by B-lymphocytes
 - (b) Provides cell mediated immunity
 - (c) Provides humoral immunity
 - (d) T-lymphocytes help to produce it
- (1) a, b, d (2) a, c, d (3) b, c (4) c, d

93. Choose the correct statement from given below.

- (a) Colostrum contains abundant IgA
- (b) Injection of dead/inactivated pathogen causes active immunity.
- (c) If due to some reason, T lymphocyte are damaged, the body will not produce antibody against pathogen.
- (d) Injection of snake antivenom against snake bite is an example of passive immunity.

- (1) a, b, d only (2) a, d only
- (3) b, c, d only (4) a, b, c only

94. Virus infected cells secrete protein calledA.... which stimulate non infected cells to produceB.... .

	(A)	(B)
(1)	Antigen	Antibody
(2)	Translation inhibiting protein	Interferon
(3)	Antibody	Antigen
(4)	Interferon	Translation inhibiting protein

95. Which of the following is correct about Natural killer cells.

- (i) Produced in bone marrow.
- (ii) Contain perforin protein
- (iii) Large agranular lymphocyte
- (iv) Non specific in function
- (v) Phagocytosis

- (1) i, ii, iv, v (2) ii, iii, v
- (3) iii, v (4) i, ii, iv

96. Read the following statements (a-d):-

- (a) Sequence of inflammatory events is Rubor, Calor, Tumor, Dolor
- (b) cell mediated immunity is responsible for graft rejection.
- (c) Best HLA matching order is Twin > Sibling > Parent > unrelated donor.
- (d) Monocyte are considered into first line of defense in immunity.

How many statements are correct.

- (1) one (2) two (3) three (4) four

97. Select the correct match :-

- (1) **Malaria** - Starting phase in human is gametocytes
- (2) **Amoebiasis** - Caused by a parasite of small intestine
- (3) **Ascariasis** - Internal bleeding, muscular pain fever and anaemia
- (4) **Elephantiasis** - It's pathogen live in blood vessels of lower limbs

98. Correct about vaccine and vaccination is :-
 (a) Suspension of antigenic protein of pathogen
 (b) Suspension of attenuated pathogens
 (c) Production of T-memory and B-memory cells
 (d) Activation of NK cells
 (e) Provide passive immunity
 (1) a + b + c + d (2) b + c + d + e
 (3) a + b + c (4) d + e
99. Select the incorrect statement for active immunity :-
 (1) When a host is exposed to antigen active immunity is stimulated
 (2) Active immunity is slow and take time to give its effectiveness
 (3) The yellowish fluid colostrum can induce the active immunity
 (4) Active immunity can be stimulated by clinical and subclinical infection.
100. At least how many afferent and efferent neurons are involved in any reflex action ?
 (1) One afferent , one efferent
 (2) One afferent, one efferent and one interneuron
 (3) One afferent, one efferent and two interneuron
 (4) Two afferent, two efferent and one interneuron
101. Which of the following hormone is not secreted by adenohypophysis ?
 (1) GH (2) TSH
 (3) ACTH (4) Oxytocin
102. Find out the correct match between column I & II:-

Column-I	Column-II
(a) FSH	(i) Darkness
(b) Thyroxine	(ii) Release of ovum
(c) MSH	(iii) Regulation of minimum O_2 consumption during resting stage
(d) LH	(iv) Regulation of sertoli cells

- (1) a – (i); b – (iii); (c) – (iv); d – (ii)
 (2) a – (iii); b – (ii); (c) – (i); d – (iv)
 (3) a – (iii); b – (iv); (c) – (ii); d – (i)
 (4) a – (iv); b – (iii); (c) – (i); d – (ii)

103. The inner ear contains a complex system located above the cochlea called(A)..... It is influenced by gravity and movements and help us in(B).....
 (1) A - organ of corti, B - Hearing
 (2) A - organ of corti, B - Maintaining posture
 (3) A - Vestibular apparatus, B - Maintaining posture
 (4) A - Macula, B - Hearing
104. Which of the following statement is false :-
 (1) Macrophages are large irregular shaped cells that engulf microbes, cellular debris in response to an infection
 (2) Acquired immunity is termed as adaptive immunity
 (3) Active immunity is used for treatment of acute infection
 (4) Local response of living mammalian tissue to injury is termed as inflammation.
105. Which of the following statement is false :-
 (1) *Salmonella typhi* causes intermittent fever
 (2) Rhino virus infect the nose and respiratory passage but not the lungs
 (3) Malarial parasite requires two hosts human and mosquitoes to complete it's life cycle
 (4) Mary mallon was a typhoid carrier and continued to spread typhoid for several years
106. Read the following statements.
 A. The atrial wall of the heart produce atrial natriuretic factor which decreases the blood pressure.
 B. Pars nervosa produce two hormone while pars distalis produces six trophic hormones.
 C. Thyroid gland Hormones (Thyroxine) play an important role in development & maturation of CNS.
 D. The cerebrum is longitudinally divided into two halves that are connected by the corpus callosum.
 How many of above statements are correct :-
 (1) Two (2) One (3) Three (4) Four

107. Choose the incorrect statement.

- (1) Endocrine control is integrated with neural control at the level of the Hypothalamus.
- (2) Thyroid hormone is required for development of CNS.
- (3) Glucocorticoids are anabolic steroids.
- (4) Oxytocin is also known as milk let down hormone.

108. Hormone that mainly acts on hepatocytes to stimulate glycogenolysis and gluconeogenesis:

- (1) Glucagon
- (2) Insulin
- (3) Prolactin
- (4) Growth hormone and oxytocin

109. Find out the correct match of following hormone with their source and functions ?

	Hormone	Source	Function
(1)	Adrenaline	Mesodermal renal part	Increase glucose level in blood & stimulate protein synthesis
(2)	Cortisol	Adrenal fasciculata layer	Stimulates lipolysis and gluconeogenesis
(3)	Thyroxine	Ectodermal thyroid	Hyperthyroidism during pregnancy causes defective development of growing baby
(4)	Aldosterone	Zona Reticularis	Maintenance of osmotic pressure

110. Which of the following hormones is related with generation of secondary messenger ?

- (1) Adrenalin and thyroxine
- (2) Estrogen and Iodothyronin
- (3) T_3 and insulin
- (4) Adrenaline and FSH

111. Glucocorticoids :-

- (1) Favours protein synthesis
- (2) Enhances Ag-Ab reactions
- (3) Secretion increases during inflammation
- (4) Tends to lower BP

112. Which of the following pair is correctly matched ?

- (1) Myoepithelial muscle contraction = Prolactin
- (2) $\uparrow$ Lipogenesis in adipose cell = Insulin
- (3) $\uparrow$ Secretion of para thyroid follicle = Thyroxine
- (4) $\uparrow$ Synthesis of muscle structural proteins - cortisol

113. Ear dust found in :-

- (1) Semi circular canal (2) Cochlea
- (3) Sacculus (4) Middle ear

114. In the given list how many structures are not included in vestibular apparatus:-

- otolith organ, organ of corti, tectorial membrane, semi circular canals, basilar membrane
- (1) 3 (2) 4 (3) 5 (4) 2

115. Which of the following drug is obtained from *claviceps purpurea* -

- (1) Charas (2) Cocaine
- (3) LSD (4) Morphine

116. Which of the following is not caused by Alcohol drinking:-

- (1) Gastritis (2) Psychosis
- (3) Emphysema (4) Loss of memory

117. In the given list how many hormones are not synthesized in pituitary gland:-

- Prolactin, FSH, Oxytocin, Vasopressin, Gonadotrophin releasing hormone(GnRH)
- (1) 5 (2) 4 (3) 2 (4) 3

118. Formation of haemozoin during life cycle of plasmodium takes place in-

- (1) Mosquito gut
- (2) Liver of human
- (3) RBC of human
- (4) Salivary gland of mosquito

119. In which of the following disease perforation of intestine may occur in severe cases:-

- (1) Filariasis (2) Typhoid
- (3) Malaria (4) Elephantiasis

120. A drug addict who is habitual of taking drug on regular basis suddenly discontinue that drug because of financial problem and shows shakiness, nausea, anxiety. These are manifestations of.....

- (1) Addiction syndrome
- (2) Withdrawal syndrome
- (3) Opioid syndrome
- (4) Immunodeficiency syndrome

121. Which of the following option is completely not true

	Column-I	Column-II	Column-III
(1)	Opioid compound	receptors present in CNS and GIT	sedative action
(2)	Cannabinoid compound	found in marijuana and charas	effect on CVS
(3)	Cocaine	inhibits dopamine transport	Energy and euphoria
(4)	Benzodiazepines	CNS stimulant	Anti sleep pills

122. Which of the following cranial nerves is not sensory in nature:-

- (1) Olfactory (2) Optic
(3) Auditory (4) oculomotor

123. The ___A___ pathway comprises at least one ___B___ neuron (receptor) and one ___C___ neuron (effector) appropriately arranged in a series. Identify A,B,C from the given options:-

- | A | B | C |
|------------|----------|----------|
| (1) reflex | efferent | afferent |
| (2) reflex | afferent | efferent |
| (3) reflex | motor | sensory |
| (4) motor | efferent | afferent |

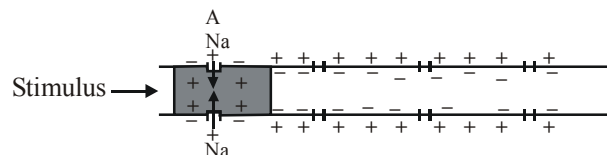
124. Reflex action :-

- (A) With conscious effort or thought and requires involuntment of a part of CNS.
(B) Reflex pathway contains at least two afferent and one efferent neuron.
(C) In reflex action first afferent neuron receive signal from sensory organ & transmit this impulse via centrifugal conduction to CNS.
(D) Sudden withdrawal of body part from object which comes in contact may include in reflex action

How many of above statements are false :-

- (1) 4 (2) 3 (3) 2 (4) 1

125. Find out the incorrect statement about the part-A of a neuron representing impulse generation in the given figure.



- (1) Membrane is freely permeable to Na⁺, which leads to rapid influx of Na⁺ ions
(2) High permeability for Na⁺ is extremely short lived
(3) Rapid influx of Na⁺ through voltage gated channels by active transport
(4) Generation of high potential called action potential

126. Which of the following is correct matching :-

(A)	<i>Wuchereria bancrofti</i>	(i)	House fly
(B)	Common round worm	(ii)	Female <i>anopheles</i>
(C)	<i>Entamoeba histolytica</i>	(iii)	Female culex
(D)	<i>Plasmodium falciparum</i>	(iv)	Contaminated food & water

	A	B	C	D
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iii)	(iv)	(ii)	(i)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(iii)	(ii)	(iv)	(i)

127. Which of the following is most feared property of malignant tumours :-

- (a) Oncogenic transformation
(b) Absence of contact inhibition
(c) Metastasis
(1) a only (2) a, b only
(3) c only (4) a, b, c

128. After getting into the body of the person HIV virus enters into macrophages where RNA genome of the virus forms viral DNA with the help of :-

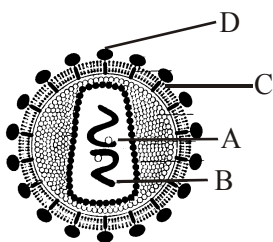
- (1) DNA polymerase (2) Transcriptase
(3) Reverse transcriptase (4) Protease

129. How many of the following statements are correct about given diagram -



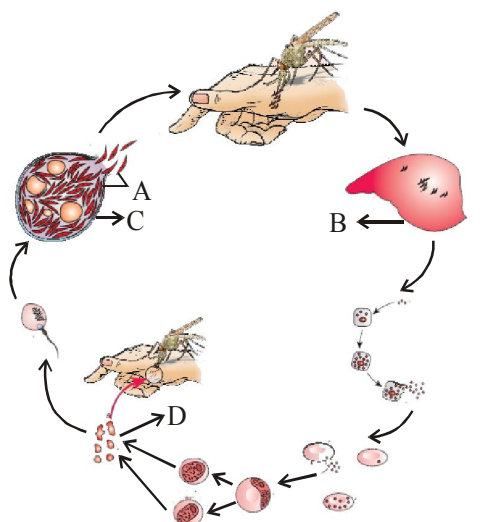
- (A) *Microsporum*, *Trichophyton* and *Epidermophyton* are responsible for this
 (B) Allergic disease
 (C) Heat & moisture help these to grow
 (D) It is caused by *Wuchereria bancrofti*
 (1) Four (2) One (3) Two (4) Three

130. Following diagram of HIV shows A, B, C and D (respectively)



- (1) SS-RNA, Reverse transcriptase, lipoprotein GP-120
 (2) GP-120, SS-RNA, Reverse transcriptase, lipoprotein
 (3) Reverse transcriptase, SS-RNA, Lipoprotein GP-120
 (4) GP-120, Lipoprotein, SS-RNA, Reverse transcriptase

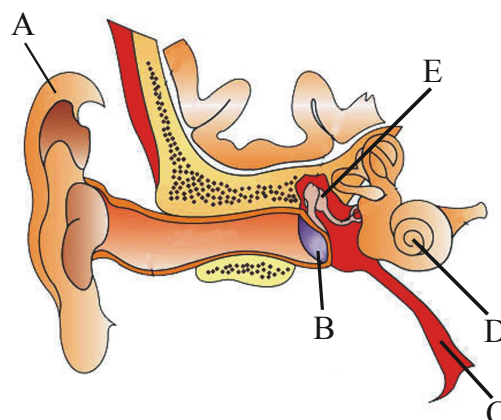
131. Identify the likely A, B, C, D in the following life cycle :-



Option

	(A)	(B)	(C)	(D)
(1)	Gametocyte	Salivary gland	Liver	Sporozoites
(2)	Sporozoites	Salivary gland	Liver	Gametocyte
(3)	Sporozoites	Liver	Salivary gland	Gametocyte
(4)	Sporozoites	Liver	Gametocytes	Salivary gland

132. Given below is a diagrammatic view of your ear. Select the option in which the structure labelled as A, B, C, D and E are correctly explained :-



- (1) A – Collects the vibration in the air which produce sound
 C – Connects middle ear cavity with bony labyrinth
 (2) A – Compose of connective tissue covered with skin outside and with mucus membrane inside
 D – Its swollen base is called ampulla which contains a projecting ridge called crista ampullaris
 (3) C – Helps in equalising the pressure on either sides of the ear drums.
 E – Increase the efficiency of transmission of sound waves on the internal ear.
 (4) B – Vibrates in response to the sound waves.
 D – Related with both balancing and hearings.

133. When ciliary muscle is _____ and suspensory ligaments becomes _____ the lens is pulled into _____ shape suitable to see distantly placed object. Which would be the most appropriate option for above situation ?

- (1) Relaxed, Loose, Concave (less convex)
- (2) Contracted, Tense, Convex
- (3) Contracted, Loose, Convex
- (4) Relaxed, Tense, less convex

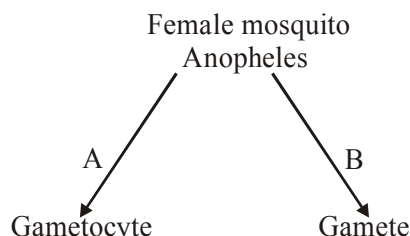
134. Which one of the following option gives the correct categorisation of hormone according to their action :-

	A Diabetogenic	B Immunity	C Erythropoiesis
1.	Cortisol Insulin GH	Gonadocorticoid T ₄	Erythropoietin Glucagon
2.	GH Aldosterone Glucagon	T ₄ Thymosin	Cortisol Androgen
3.	T ₄ Glucagon Adrenaline	Cortisol Thymosin	T ₄ Androgen
4.	GH Glucocorticoid Glucagon	Thymosin Aldosterom	Erythropoietin Cortisol

135. Which of following statement is false :-

- (1) Organ transplantation between monozygotic twin does not require MHC matching.
- (2) Avascular cardiac valve transplant from other species animal (like pig) is possible while avascular corneal transplant is not.
- (3) MHC gene hereditically transfer from only single parents
- (4) Organ transplantation between sibling have more or less chances than 50% while from parents it is not more than 50% commonly

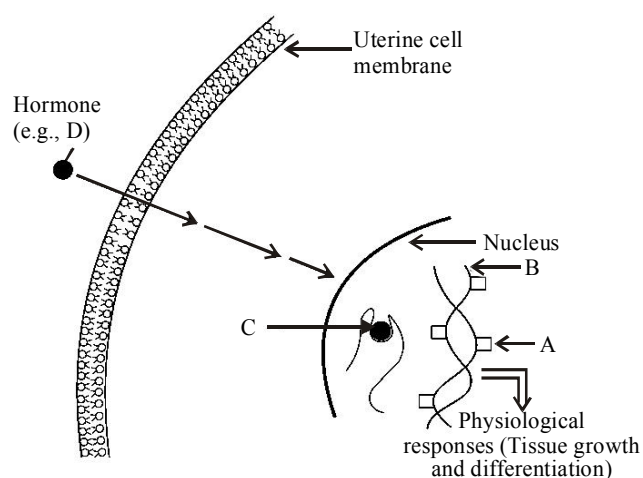
136.



In above diagram A and B are formed in respectively :-

- (1) Gut of mosquito & gut of human.
- (2) RBC of mosquito & RBC of human.
- (3) RBC of human & gut of mosquito.
- (4) Gut of human & RBC of mosquito.

137. Find out the correct labelling in the following diagram?



- (1) A = Glucose, B = Genome, C = Receptor, D = Adrenaline
- (2) A = Fats, B = DNA, C = Hormone Receptor complex, D = Thyroxine
- (3) A = Proteins, B = DNA, C = Hormone Receptor complex, D = Adrenaline
- (4) A = Proteins, B = Genome, C = Hormone Receptor complex, D = Oestrogen

138. Due to hyposecretion of an endocrine gland disease occur :-

- (1) Gigantism
- (2) Grave's disease
- (3) Addison's disease
- (4) Acromegally

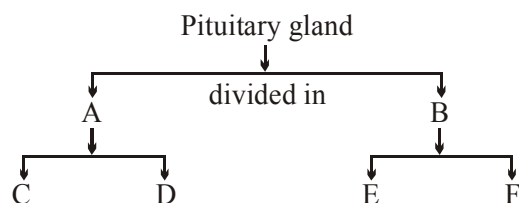
139. Anterior lobe of pituitary gland secretes :-

- (A) FSH, GH and LH
- (B) STH, ACTH and TSH
- (C) FSH, ADH and prolactin
- (D) ACTH, TSH and oxytocin

Which option is correct :-

- (1) A and B are correct
- (2) B and D are correct
- (3) A and C are correct
- (4) A, B and C are correct

140.



In above diagram F is hormone which contract uterine muscle. Then which hormone is E

- (1) Pitocin
- (2) Pitressin
- (3) ACTH
- (4) MSH

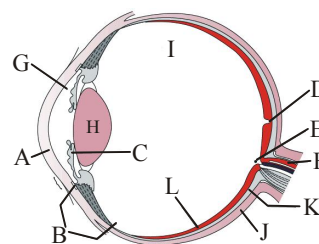
141. Which is not an example of fat soluble hormone:-

- (1) Estrogen
- (2) Thyroxine
- (3) Testosterone
- (4) Prolactin

142. ____ (I) ____ is a structure which contains hair cells that act as ____ (II) ____ and is located on the ____ (III) ____.

	(I)	(II)	(III)
(1)	vestibular apparatus	balancing organ	reissner's membrane
(2)	organ of corti	Hearing organ	Tectorial membrane
(3)	organ of corti	auditory receptors	basilar membrane
(4)	organ of corti	auditory receptors	vestibular membrane

143. Given below is diagram showing parts of an eye. Which of them is/are incorrectly described ?



- (1) A - cornea, B - ciliary body, C - iris
- (2) D - fovea, E - blind spot, F - optic nerve
- (3) G - aqueous chamber, H - Lens, I - vitreous chamber
- (4) J - retina, K - sclera, L - choroid

144. Which one is a primary sex organ ?

- (1) Scrotum
- (2) Penis
- (3) Testis
- (4) Prostate gland

145. Secondary sex organ is

- (1) Testis
- (2) Ovary
- (3) Testis & Ovary
- (4) Vas deferens

146. Which acid occur in semen ?

- (1) Citric acid
- (2) Malic acid
- (4) Oxalo acetic acid
- (4) Succinic acid

147. Vasa deferens is cut for

- (1) Female sterilization
- (2) Male sterilization
- (3) Both of the above
- (4) Temporary sterilization

148. The function of seminal fluid is -

- (1) sexual attraction
- (2) to provide stability to egg
- (3) to provide a medium for the movement of sperms
- (4) to provide acidic medium

149. Vasa efferentia connect the

- (1) testes with epididymis
- (2) kidneys with cloaca
- (3) Testes with urinogenital duct
- (4) All of the above

150. In mammals, failure of testes to descend into scrotum is known as

- (1) Paedogenesis
- (2) Castration
- (3) Cryptorchidism
- (4) Impotency

- 151.** Common duct formed by union of vas deferens and duct from seminal vesicle is
 (1) urethra (2) tunica-vasculosa
 (3) ejaculatory duct (4) spermatic duct
- 152.** Scrotum communicates with abdominal cavity through
 (1) urethra (2) inguinal canal
 (3) vas deferens (4) epididymis
- 153.** Tunica albuginea is the covering around
 (1) Oviduct (2) Testis
 (3) Kidney (4) Heart
- 154.** The functional unit of testis of man is
 (1) Uriniferous tubules
 (2) Malpighian tubules
 (3) Seminiferous tubules
 (4) Acini or lobules
- 155.** Penile urethra traverses through
 (1) Corpora cavernosa
 (2) Corpus spongiosum
 (3) Corpus callosum
 (4) Corpus striatum
- 156.** Seminiferous tubules are composed of
 (1) Periotoneum
 (2) Glandular epithelium
 (3) Sensory epithelium
 (4) Germinal epithelium
- 157.** In humans, the testes are located in
 (1) Abdominal cavity
 (2) Thoracic cavity
 (3) Extra-abdominal scrotum
 (4) Pericardial cavity
- 158.** Sustentacular cells are found in
 (1) Testis of mammal
 (2) Ovary of mammal
 (3) Testis of *ascaris*
 (4) Pancreas of frog

- 159.** Clitoris is homologous to :-
 (1) Penis
 (2) Prostate gland
 (3) Breast
 (4) Vagina
- 160.** Mesoderm derived tissue absent in :-
 (1) *Pavo cristatus* (2) *Hyla*
 (3) *Obelia* (4) *Trygon*
- 161.** Find the correct match :-

	Column I	Column II	Column III
	Animal	Class	Kidney
(1)	<i>Naja</i>	Chordata	Metanephric
(2)	<i>Macropus</i>	Mammalia	Metanephric
(3)	<i>Labeo</i>	Chondrichthyes	Metanephric
(4)	<i>Corvus</i>	Aves	Mesonephric

- 162.** Find the incorrect match :-

	Column I	Column II
	Macrophage	Tissue/Organ
(1)	Dust cell	Lungs
(2)	Microglial cell	Nervous tissue
(3)	Kupffer cell	Liver
(4)	Histiocyte	Bone

- 163.** Read the following structures :-
 Bone, Cartilage, Tendon, Ligament, Adipose tissue, Hairs, Biceps, Nasal septum.
 How many above cited structures can be included in connective tissue.
 (1) 8 (2) 6
 (3) 4 (4) 3
- 164.** Time difference between closing of aortic semilunar valve and mitral valve respectively is :-
 (1) 0.3 sec (2) 0.5 sec
 (3) 0.1 sec (4) 0.8 sec

165. Conduction system of human heart is :-

- (1) Modified smooth muscles
- (2) Modified skeletal muscles
- (3) Modified cardiac muscles
- (4) Modified nervous tissue

166. Coronary sinus open in :-

- (1) Left atrium of heart
- (2) Right atrium of heart
- (3) Right ventricle of heart
- (4) Left ventricle of heart

167. The blood vessels which carry the blood from the heart is :-

- (1) Vein
- (2) Artery
- (3) Venules
- (4) Capillaries

168. All of the following structures are jointed in *Periplaneta* except :-

- (1) Antennae
- (2) Legs
- (3) Labial palp
- (4) Anal style

169. Which of the following is not related to tadpole larva of *Rana tigrina* ?

- (1) Gills
- (2) Ammonotelism
- (3) Carnivorous
- (4) Pronephric kidney

170. Which of the following functional gene is absent in human genome ?

- (1) Gene which code pepsinogen
- (2) Gene which code maltase
- (3) Gene which code nucleosidase
- (4) Gene which code cellulase

171. All of the following are part of human digestive system except :-

- (1) Enamel, crypts of Lieberkuhn
- (2) Meissner plexus, Brunner's gland
- (3) Gastric glands, caecum
- (4) Cysterna chyli, subclavian vein

172. Endomysium is present around :-

- (1) A muscle
- (2) A fasciculus
- (3) A muscle fibre
- (4) A sarcomere

173. Which bone of leg is a sesamoid bone ?

- (1) Femur
- (2) Calcaneum
- (3) Patella
- (4) Talus

174. Epiphysis of femur is covered by :-

- (1) Hyaline cartilage
- (2) White fibrous cartilage
- (3) Elastic cartilage
- (4) Stratified squamous keratinized epithelium

175. Find the correct match :-

	Column I	Column II	Column III
(1)	PCT	Brush border simple columnar epithelium	Maximum tubular secretion
(2)	Jejunum	Brush border simple cuboidal epithelium	Maximum absorption of digested food
(3)	Left ventricle	Thickest myocardium	Pump the blood into systemic aorta
(4)	Radius	Bone of arm	Thicker and stronger than ulna

176. The total amount of blood supply to both kidney in human per minute is (under normal physiological condition) :-

- (1) 150-200 ml
- (2) 650-700 ml
- (3) 20% to 22% of total cardiac output
- (4) 2000 ml to 3000 ml

177. Maximum tubular reabsorption of Na⁺ take place in :-

- (1) PCT
- (2) DCT
- (3) Loop of Henle
- (4) Collecting duct

178. Find the incorrect match :-

- (1) Oxyntic cell – HCl
- (2) JG cell – Erythropoietin
- (3) Hepatocytes – Synthesis of bile juice
- (4) Osteoclast – Bone forming cells

179. Find the incorrect match :-

- (1) *Balaenoptera* → Mammary gland →
Modified sweat gland
- (2) *Bufo* → 3 chambered heart →
Pump mixed blood
- (3) *Ascaris* → Notochord absent →
Invertebrate nonchordate
- (4) *Physalia* → Colloblast → Stinging cell

180. Which part of nephron has receptor for vasopressin :-

- (1) Bowman capsule
- (2) Proximal convoluted tubule
- (3) Distal convoluted tubules
- (4) Duct of Bellini



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ANSWER KEY
PRE-MEDICAL 2020

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	4	4	3	2	1	2	3	1	3	4	4	1	2	1	1	2	1	3	4	3	3	4	3	2	2	1	1	3	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	1	1	2	3	2	2	3	4	4	2	3	1	3	3	2	1	2	3	2	2	4	1	4	3	1	3	2	1	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	3	1	4	1	2	2	2	3	2	4	4	4	3	4	1	3	3	4	3	3	4	3	1	1	3	2	3	2	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	2	1	4	4	3	3	3	3	1	4	4	3	3	1	3	3	1	2	4	3	2	3	1	3	3	4	3	2	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	2	2	3	3	3	3	3	3	3	3	4	3	3	3	4	3	1	2	4	3	4	3	4	1	2	3	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	2	2	3	2	4	3	1	1	3	2	4	2	2	3	2	2	4	3	4	4	3	3	1	3	3	1	4	4	3

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TEST DATE : 16 - 02 - 2020

HINT - SHEET

1. If the sheet is heated then both d_1 and d_2 will increase since the thermal expansion of isotropic solid is similar to true photographic enlargement.

2. Change in length for both rods should be same

$$\Delta l_1 = \Delta l_2$$

$$l_1 \alpha_1 \Delta T = l_2 \alpha_2 \Delta T$$

$$l_1 \alpha_1 = l_2 \alpha_2$$

3. At 0°C from b to c, temperature of matter does not change but its state changes.

Similarly, from d to e state of matter changes without changing temperature. Hence, b to c and d to e show state changes.

4. We are given the relation between the

$$\text{temperature scale A and B as } \frac{A-42}{110} = \frac{B-72}{220},$$

so for the two scales to have the same reading, $A = B$ and so,

$$\frac{A-42}{110} = \frac{A-72}{220}$$

This gives $A = 12$

5. Let the final temperature be $T^\circ\text{C}$

Total heat supplied by the three liquids in coming down to 0°C

$$= m_1 c_1 T_1 + m_2 c_2 T_2 + m_3 c_3 T_3 \quad \dots(1)$$

Total heat used by three liquids in raising temperature from 0°C to $T^\circ\text{C}$.

$$= m_1 c_1 T + m_2 c_2 T + m_3 c_3 T \quad \dots(2)$$

By equating Eqs. (i) and (ii), we get

$$(m_1 c_1 + m_2 c_2 + m_3 c_3) T = m_1 c_1 T_1 + m_2 c_2 T_2 +$$

$$m_3 c_3 T_3 \Rightarrow T = \frac{m_1 c_1 T_1 + m_2 c_2 T_2 + m_3 c_3 T_3}{m_1 c_1 + m_2 c_2 + m_3 c_3}$$

6. Let temperature of interface is

$$T \Rightarrow \left(\frac{dQ}{dt} \right)_1 = \left(\frac{dQ}{dt} \right)_2$$

$$\frac{KA(300-T)}{l} = \frac{2KA(T-0)}{l} \Rightarrow T = 100^\circ\text{C}$$

7. Efficiency of a Carnot engine

$$\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{(273+27)}{(273+127)} = 1 - \frac{300}{400} = \frac{1}{4}$$

$$\eta = \frac{\text{Work done}}{\text{Heat supplied}}$$

$$\Rightarrow \frac{1}{4} = \frac{W}{40} \Rightarrow W = 10 \text{ kJ}$$

8. For mixture of gases, let specific heat be C_V

$$C_V = \frac{n_1(C_V)_1 + n_2(C_V)_2}{n_1 + n_2}$$

$$\text{Where } (C_V)_1 = \frac{5R}{2}, (C_V)_2 = \frac{3R}{2}$$

$$= \frac{2 \times \frac{5R}{2} + 8 \times \frac{3R}{2}}{2+8} = \frac{17R}{10} = 1.7R$$

$$9. v_{\text{rms}} = \sqrt{\frac{(3v)^2 + (4v)^2 + (5v)^2}{3}} = \sqrt{\frac{50}{3}} v$$

10. From Newton's law of cooling

$$\frac{\theta_1 - \theta_2}{t} \propto \left(\frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$$

Therefore,

$$\frac{60^\circ - 40^\circ}{7} \propto \left(\frac{60^\circ + 40^\circ}{2} - 10^\circ \right) \quad \dots\dots(i)$$

$$\frac{40^\circ - 28^\circ}{t} \propto \left(\frac{40^\circ + 28^\circ}{2} - 10^\circ \right) \quad \dots\dots(ii)$$

On solving the Eqs. (i) and (ii), we get $t = 7\text{min}$

11. Emissive power of a body with surface temperature (TK) is given as

$$E = \sigma T^4$$

$$\text{Hence, } \frac{E_1}{E_2} = \left(\frac{T_1}{T_2} \right)^4 = \left(\frac{273 - 73}{273 + 327} \right)^4 = \left(\frac{200}{600} \right)^4$$

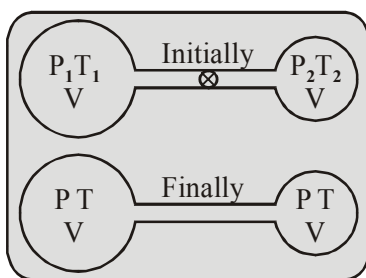
$$E_1 : E_2 = 1 : 81$$

12. Number of moles in first vessel

$$\mu_1 = \frac{P_1 V}{RT_1}$$

Number of moles in second vessel

$$\mu_2 = \frac{P_2 V}{RT_2}$$



If both vessels are joined together, then quantity of gas remains same, i.e., $\mu = \mu_1 + \mu_2$

$$\frac{P(2V)}{RT} = \frac{P_1 V}{RT_1} + \frac{P_2 V}{RT_2}$$

$$\frac{P}{T} = \frac{P_1}{2T_1} + \frac{P_2}{2T_2}$$

13. In an adiabatic process

$$PV^\gamma = \text{constant}$$

$$\Rightarrow \frac{P_1}{P_2} = \left(\frac{V_2}{V_1} \right)^\gamma \Rightarrow \frac{P_1}{P_2} = \left(\frac{1}{8} \right)^{5/3}$$

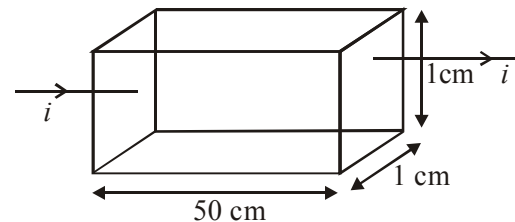
$$\Rightarrow \frac{P_1}{P_2} = \left(\frac{1}{2^3} \right)^{5/3} \Rightarrow \frac{1}{32} \Rightarrow \frac{P_2}{P_1} = 32$$

$$14. q = \int_0^{t=1} Idt = \int_0^{t=1} (1 + 2t) dt = 2C$$

$$15. J = \frac{I}{A} \text{ and } J = \sigma E$$

16.

17.



$$R = \frac{\rho L}{A} = \frac{3.5 \times 10^{-5} \times 50 \text{ cm}}{1 \text{ cm} \times 1 \text{ cm}} = 0.175 \Omega$$

$$18. V = E - ir$$

19. By stretching a wire

$$R \propto \ell^2 \quad \ell_2 = \ell_1 + 50\% \quad \ell_1 = \ell_1 + \frac{50}{100} \ell_1$$

$$\frac{R_2}{R_1} = \left(\frac{\ell_2}{\ell_1} \right)^2 \quad \ell_2 = \frac{3}{2} \ell_1$$

$$\frac{R_2}{R_1} = \left(\frac{3}{2} \right)^2 \quad \frac{\ell_2}{\ell_1} = \frac{3}{2}$$

$$\frac{R_2}{R_1} = \frac{9}{4}$$

$$\frac{R_2}{R_1} - 1 = \frac{9}{4} - 1$$

$$\frac{R_2 - R_1}{R_1} = \frac{9 - 4}{4} \times 100 = 125\%$$

20. $A \xrightarrow{3A} \xrightarrow{+2\Omega} \xrightarrow{5V} \xrightarrow{+1\Omega} B$
 $6V \quad 3V$

$$V_A - 6 + 5 - 3 = V_B \Rightarrow$$

$$V_B - V_A = -4 \text{ volt}$$

21. $V_A - V_B = \frac{20(-5) + 10(2)}{10 + 20}$

$$V_A - V_B = -\frac{8}{3} \text{ volt}$$

22. At surface of earth,

$$g = \frac{GM}{R^2} \quad \dots(i)$$

At a height, $h = R/2$, the value of acceleration due to gravity is

$$g' = \frac{GM}{(h+R)^2} = \frac{GM}{\left(\frac{R}{2} + R\right)^2} = \frac{GM}{\left(\frac{3R}{2}\right)^2} = \frac{GM}{\left(\frac{3R}{2}\right)^2},$$

$$g' = \frac{4}{9} \frac{GM}{R^2} = \frac{4}{9} g$$

$\therefore$ weight at a height $h = R/2$ is

$$W' = mg' = \frac{4}{9} mg = \frac{4}{9} W.$$

23. $g = \left(\frac{GM_e}{R_e^3}\right)r$ for $0 < r \leq R_e \Rightarrow g \propto r$

$$g = \frac{GM_e}{r^2} \quad \text{for } r \geq R_e \Rightarrow g \propto \frac{1}{r^2}$$

24. $\frac{\Delta g}{g} = \frac{2h}{R}, \quad \frac{\Delta g}{g} = \frac{d}{R}$

$$\text{So } \frac{2h}{R} = \frac{d}{R}$$

$$2 \times 1 = d$$

$$d = 2 \text{ km}$$

25. $12 \times \ell = m \times \frac{\ell}{2} + 3 \times \left(\frac{3\ell}{2}\right)$

26. $\omega_0 = 240 \text{ rpm} = 240 \times \frac{2\pi}{60} = 8\pi \text{ rad/sec}$

$$\omega = \omega_0 + \alpha t$$

$$0 = 8\pi - \alpha(60) \Rightarrow \alpha = \frac{8\pi}{60} = \frac{2\pi}{15} \text{ rad/s}^2$$

$$\tau = I\alpha = 4 \left(\frac{2\pi}{15}\right) = \frac{8\pi}{15} \text{ Nm}$$

27. $I_1 = \frac{2}{5} MR^2, \quad I_2 = \frac{2}{5} M \left(\frac{R}{2}\right)^2 = \frac{I_1}{4}$

$$I\omega = \text{constant}$$

$$I \times \frac{2\pi}{T} = \text{constant} \Rightarrow I \propto T$$

28. $\mu \geq \frac{\tan \theta}{1 + R^2 / K^2}$

29. For sliding $t_1 = \frac{1}{\sin \theta} \sqrt{\frac{2h}{g}}$

For rolling t_2

$$= \frac{1}{\sin \theta} \sqrt{\frac{2h}{g} \left(1 + \frac{K^2}{R^2}\right)} = \frac{1}{\sin \theta} \sqrt{\frac{2h}{g} (1+1)}$$

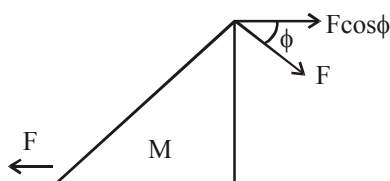
$$= \frac{1}{\sin \theta} \sqrt{\frac{4h}{g}}$$

30. $F - \mu_k mg = ma$

$$40 - \mu_k (10 \times 10) = 10 \times 2$$

$$\Rightarrow \mu_k = 0.2$$

31. $a = \frac{F_{\text{net}}}{M} = \frac{F - F \cos \phi}{M}$

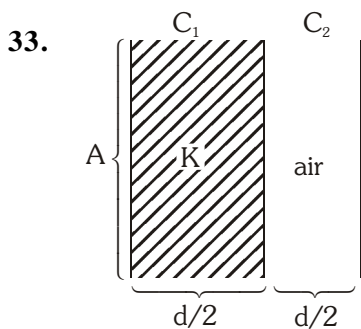


32. C and C are in parallel and in series with 2C.
Therefore resultant of these three will be

$$= \frac{(C+C) \times 2C}{C+C+2C} = C$$

This equivalent system is in parallel with C.
It's equivalent capacitance = C + C = 2C
Now, 2C and 2C are in series and in parallel with 2C.

$$\text{So, } C_{\text{net}} = C + 2C = 3C$$



$$C = \frac{\epsilon_0 A}{d}$$

$$C_1 = \frac{\epsilon_0 KA}{d/2} = \frac{2\epsilon_0 KA}{d}$$

$$C_2 = \frac{\epsilon_0 A}{d/2} = 2\left(\frac{\epsilon_0 A}{d}\right)$$

C_1 & C_2 are in series combination
Hence

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d}{2\epsilon_0 A} \left(\frac{1}{K} + \frac{1}{1} \right)$$

$$\frac{1}{C_s} = \frac{d}{2\epsilon_0 A} \left(\frac{K+1}{K} \right)$$

$$C_s = \frac{2\epsilon_0 A}{d} \left(\frac{K}{K+1} \right)$$

$$C_s = \frac{2CK}{K+1}$$

34. Charge in two spheres will be same. Let V_1 and V_2 be the potential difference for the two spheres then

$$V_1 + V_2 = V = 100 \quad \dots(i)$$

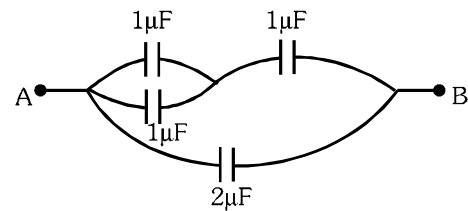
$$q_1 = q_2 \text{ and } C_1 = C; C_2 = 4C$$

$$CV_1 = 4CV_2$$

$$V_1 = 4V_2 \quad \dots(ii)$$

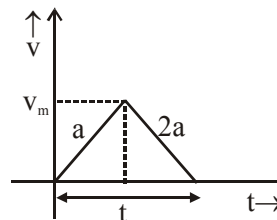
On solving $V_1 = 80 \text{ V}$ and $V_2 = 20 \text{ V}$

- 35.



Redraw the ckt. $C_{AB} = \frac{8}{3} \mu\text{F}$

- 36.



$$\text{Now, } \frac{v_m}{t_1} = a, \frac{v_m}{t_2} = 2a \text{ \& } t_1 + t_2 = t$$

solve and get

$$v = \frac{2at}{3}$$

$$\text{Now distance} = \frac{1}{2} \times t \times \frac{2at}{3} = \frac{at^2}{3}$$

37. Area of v-t graph = displacement $\approx$ disp.

Distance = 55 m